

Capture the Flag 3: Data-Driven UI

Goal: Separate application data from the UI and handle multiple pieces of data in a usable interface.

Instructor Note

These activities require your own problem-solving and research skills.

DO NOT use AI to generate the code for you.

You may use AI to:

- Research what a Dart/Flutter feature does
- Understand syntax
- **Explain** an error (not **fix** your error)
- Clarify documentation

You may **NOT** ask AI to generate the solution/code for a flag.

🚩 Flag 0 — Create Your Feature Branch

```
C:\Users\Manes Christopher\Documents\mannes\IT314AB_Artajo\my_first_flutter_app>git branch
* feature/data-ui
  feature/widgets
  main
```



Flag 1 — Identify the Data

Objective: Identify the appropriate Dart data type for the information in your profile.

Research basic Dart data types and identify at least **four different types** that could be used in your application.

Examples:

String
int
double
bool

Create a table identifying the information and its appropriate data type.

Create:

- Your identified data
- The corresponding data types

Information	Data type
name	String
Height	double
Age	int
Student Status	bool

Flag 2 — Create Your Data

Objective: Create appropriately typed variables for your profile.

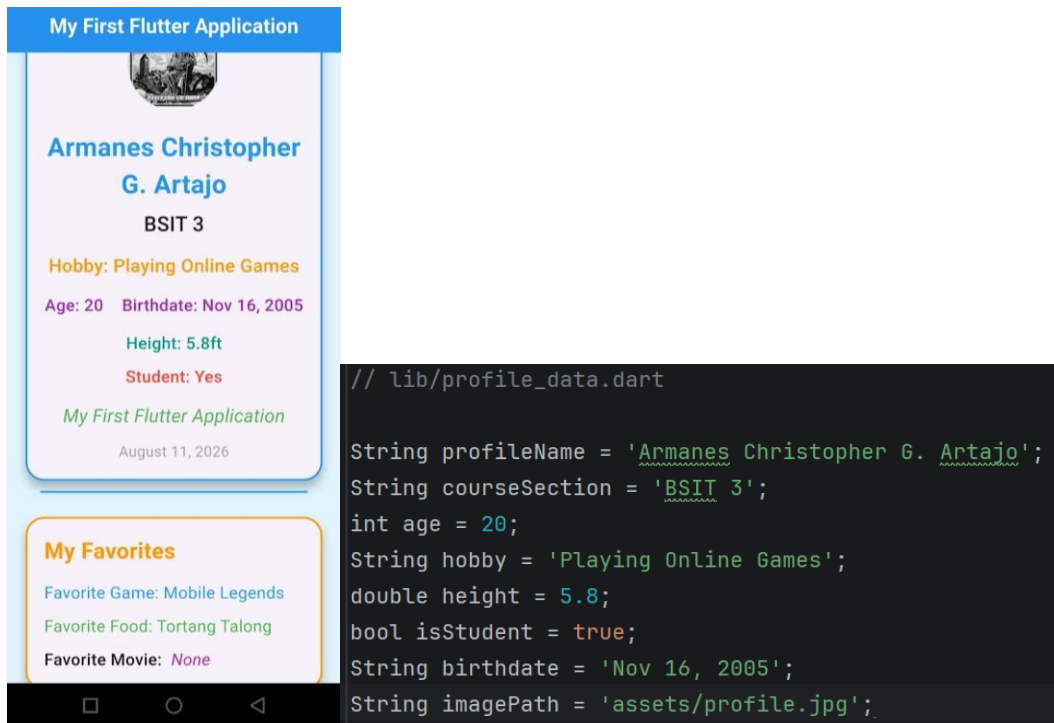
Create variables for at least:

- Name
- Course & Section
- Age
- Hobby
- Height
- Student status

Your data should exist **separately from your UI widgets**. (Declare variables separately from your UI, not directly inside your widgets)

Screenshot:

- Your variables
- Their data types
- Application still running





Flag 3 — Connect Data to UI

Objective: Replace your hardcoded information with your variables.

Your profile should now display information using the variables you created.

Test:

Change the value of one of your variables.

The UI should automatically display the new value.

Screenshot:

- Variables
- Widgets displaying the variables
- Updated application

```
String myName = 'Mannes Artajo';
```



🚩 Flag 4 — Make the Image Data-Driven

Objective: Make your profile image part of your data.

Create a variable containing the path to your profile image.

Use the variable when displaying the image.

Then change the variable to another registered image.

The image displayed by the application should change.

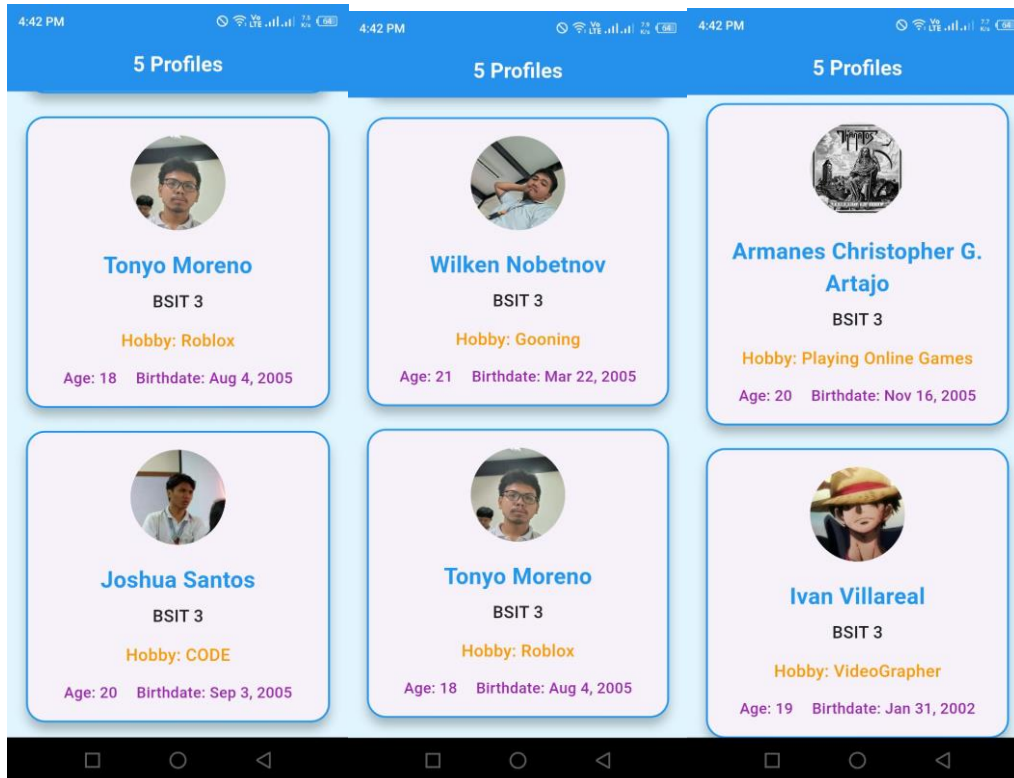
Screenshot:

- Image variable
- Image widget using the variable
- Updated application

```
String myImage = 'assets/unnamed.jpg';
```



🚩 Flag 5 — Five Profiles, One Design



```
String name1 = 'Armanes Christopher G. Artajo';
String course1 = 'BSIT 3';
int age1 = 20;
String hobby1 = 'Playing Online Games';
String image1 = 'assets/profile.jpg';
String birthdate1 = 'Nov 16, 2005';

// Profile 2
String name2 = 'Ivan Villareal';
String course2 = 'BSIT 3';
int age2 = 19;
String hobby2 = 'VideoGrapher';
String image2 = 'assets/images.jpg';
String birthdate2 = 'Jan 31, 2002';

// Profile 3
String name3 = 'Wilken Nobetnov';
String course3 = 'BSIT 3';
int age3 = 21;
String hobby3 = 'Gooning';
String image3 = 'assets/wklen.jpg';
String birthdate3 = 'Mar 22, 2005';

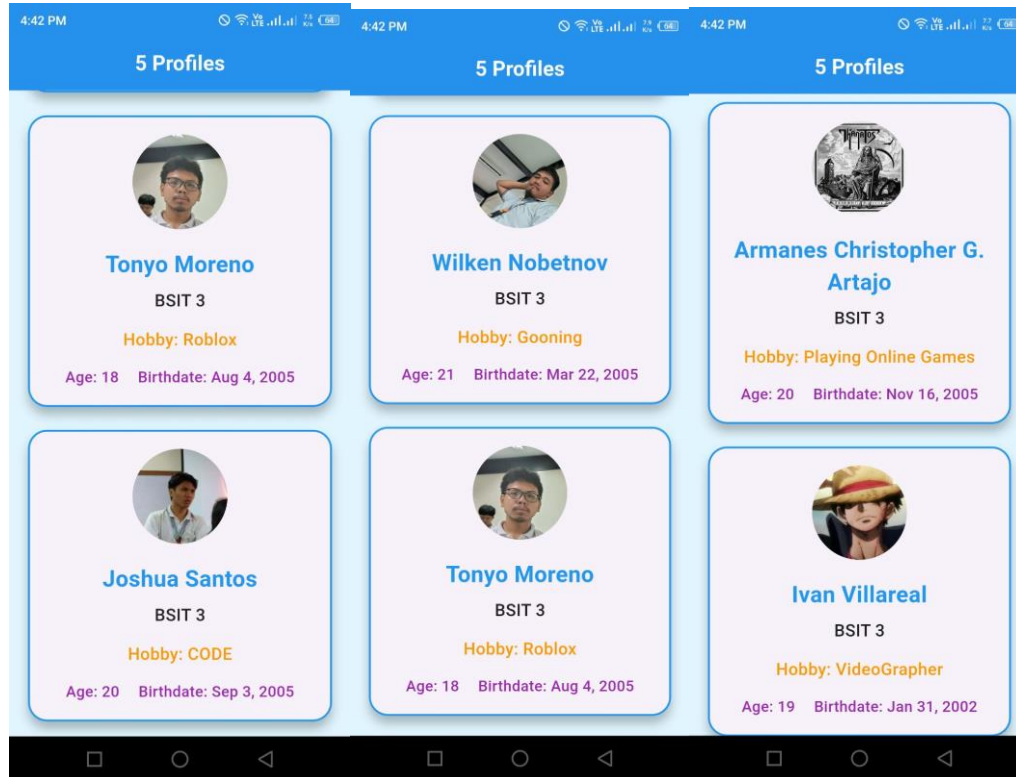
// Profile 4
String name4 = 'Tonyo Moreno';
String course4 = 'BSIT 3';
int age4 = 18;
String hobby4 = 'Roblox';
String image4 = 'assets/toning.jpg';

// Profile 5
String name5 = 'Joshua Santos';
String course5 = 'BSIT 3';
int age5 = 20;
String hobby5 = 'CODE';
String image5 = 'assets/shuuaa.jpg';
String birthdate5 = 'Sep 3, 2005';

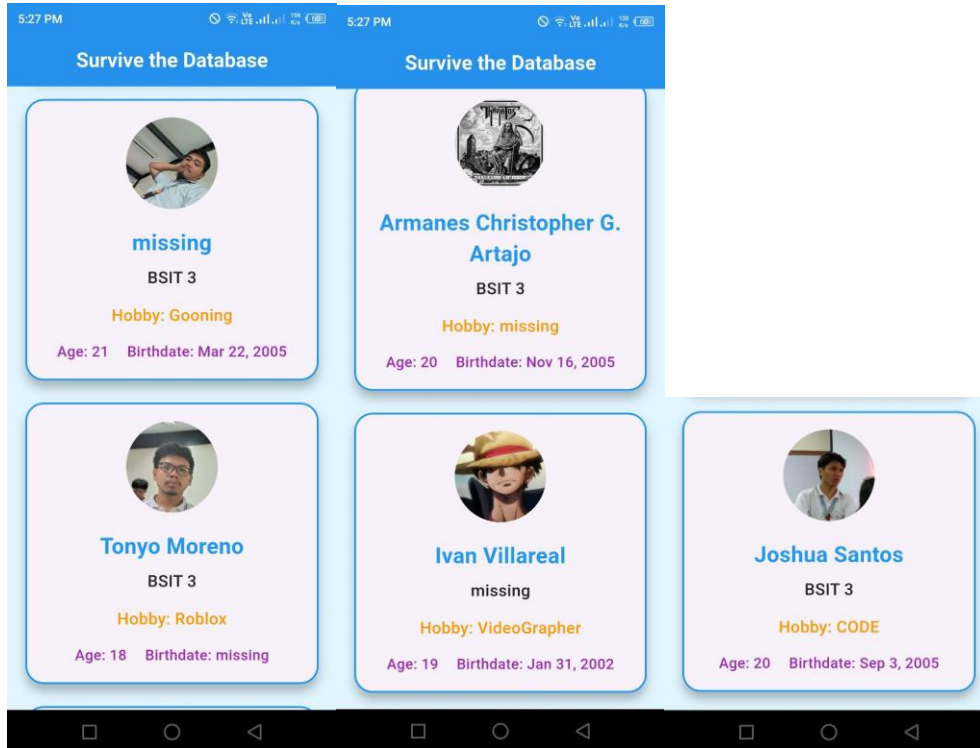
// App info (shared)
String appTitle = 'My First Flutter Application';
String todayDate = 'August 11, 2026';
```

🚩 Flag 6 — THE HIDDEN PROBLEM

```
body: SingleChildScrollView(
  padding: EdgeInsets.all(16),
  child: Column(
    mainAxisAlignment: MainAxisAlignment.min,
    children: [
```



Flag 7 — Survive the Database



```
// Profile 1: Missing hobby
String nameA = 'Armanes Christopher G. Artajo';
String courseA = 'BSIT 3';
int ageA = 20;
String? hobbyA = null; // Missing
String imageA = 'assets/profile.jpg';
String birthdateA = 'Nov 16, 2005';

// Profile 2: Missing course
String nameB = 'Ivan Villareal';
String? courseB = null; // Missing
int ageB = 19;
String hobbyB = 'VideoGrapher';
String imageB = 'assets/images.jpg';
String birthdateB = 'Jan 31, 2002';

// Profile 3: Missing name
String? nameC = null; // Missing
String courseC = 'BSIT 3';
int ageC = 21;
String hobbyC = 'Gooning';
String imageC = 'assets/wkLen.jpg';
String birthdateC = 'Mar 22, 2005';

// Profile 4: Missing birthdate
String nameD = 'Tonyo Moreno';
String courseD = 'BSIT 3';
int ageD = 18;
String hobbyD = 'Roblox';
String imageD = 'assets/toning.jpg';
String? birthdateD = null; // Missing

// Profile 5: Complete
String nameE = 'Joshua Santos';
String courseE = 'BSIT 3';
int ageE = 20;
String hobbyE = 'CODE';
String imageE = 'assets/shuuaa.jpg';
String birthdateE = 'Sep 3, 2005';

String appTitle = 'Survive the Database';
```



Flag 8 — Save Your Progress

Objective: Commit and push your completed Data-Driven UI CTF.

Use an appropriate commit message following proper Git standards.

Branch:

`feature/data-ui`

Screenshot:

- Successful commit
- Current branch
- Terminal showing the commit

```
C:\Users\Manes Christopher\Documents\mannes\IT314AB_Artajo>git commit -m "feat: complete data-driven UI CTF with profile management"
[feature/data-ui 7e4e5dc] feat: complete data-driven UI CTF with profile management
 8 files changed, 236 insertions(+), 172 deletions(-)
 create mode 100644 my_first_flutter_app/assets/images.jpg
 create mode 100644 my_first_flutter_app/assets/shuaa.jpg
 create mode 100644 my_first_flutter_app/assets/toning.jpg
 create mode 100644 my_first_flutter_app/assets/unnamed.jpg
 create mode 100644 my_first_flutter_app/assets/wklen.jpg
 create mode 100644 my_first_flutter_app/lib/profile_data.dart
```